

BS-PH101 — Physics-I

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Module 1 Comprehensive Exam Notes: Mechanics

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Constraints and Friction

1.1 Constraints and Degrees of Freedom

Definition: Constraint and Degrees of Freedom

A **constraint** is any physical restriction on the position or motion of a mechanical system. The number of independent coordinates required to completely specify the configuration of a system is called its **Degrees of Freedom (DOF)**.

If a system consists of N particles in 3D space subject to k independent constraint equations, the number of degrees of freedom is given by:

$$f = 3N - k \quad (1)$$

Classification of Constraints:

Constraint Type	Condition / Mathematical Form	Physical Example
Holonomic	Expressible as $f(r_1, r_2, \dots, r_N, t) = 0$ (integrable algebraic relations).	Rigid body ($ \vec{r}_i - \vec{r}_j = c$), simple pendulum ($r = L$).
Non-Holonomic	Expressible as inequalities or non-integrable differential relations ($f(\vec{r}, \dot{\vec{r}}, t) = 0$).	Rolling without slipping on a rough plane, gas molecules in a container ($r \leq R$).
Scleronomic	Constraint equations do not explicitly depend on time t ($\partial f / \partial t = 0$).	Bead on a fixed wire ring, rigid dumbbell.
Rheonomic	Constraint equations explicitly depend on time t ($\partial f / \partial t \neq 0$).	Bead on a wire ring rotating with constant angular velocity ω .

1.2 Friction and Laws of Friction

Friction is the resistive contact force that opposes the relative motion or tendency of relative motion between two surfaces in contact.

Key Laws and Relations of Friction

- **Static Friction (f_s):** Self-adjusting force. Maximum value is limiting friction:

$$f_{s,\max} = \mu_s N \quad (2)$$

- **Kinetic Friction (f_k):** Constant resistive force during motion:

$$f_k = \mu_k N \quad (\text{where } \mu_k < \mu_s) \quad (3)$$

- **Angle of Friction (λ):** Angle between total contact reaction $\vec{R}$ and normal reaction $\vec{N}$:

$$\tan \lambda = \frac{f_{s,\max}}{N} = \mu_s \implies \lambda = \tan^{-1}(\mu_s) \quad (4)$$

- **Angle of Repose (θ_r):** Maximum angle of inclination of a rough plane at which a body remains at rest without sliding:

$$mg \sin \theta_r = \mu_s mg \cos \theta_r \implies \tan \theta_r = \mu_s \implies \theta_r = \lambda \quad (5)$$

Solved Problem 1: Block on an Inclined Plane with Friction

Problem: A block of mass $m = 5$ kg rests on an inclined plane of angle $\theta = 30^\circ$. The coefficient of static friction is $\mu_s = 0.4$. Determine whether the block slides down, and find the frictional force.

Solution:

1. Component of gravity along the incline: $F_{\text{down}} = mg \sin 30^\circ = 5 \times 9.8 \times 0.5 = 24.5$ N.
2. Normal reaction: $N = mg \cos 30^\circ = 5 \times 9.8 \times 0.866 = 42.435$ N.
3. Maximum limiting static friction: $f_{s,\text{max}} = \mu_s N = 0.4 \times 42.435 = 16.97$ N.
4. Since $F_{\text{down}}(24.5 \text{ N}) > f_{s,\text{max}}(16.97 \text{ N})$, the block **will slide down**. The friction force acting up the plane during motion is kinetic friction $f_k = \mu_k N$.

Solved Problem 2: Degrees of Freedom of Connected Bodies

Problem: Find the degrees of freedom for: (a) A double pendulum moving in a plane, (b) Two particles connected by a rigid rod free to move in 3D space.

Solution:

1. **Double Pendulum in 2D:** $N = 2$ particles, constrained to plane ($z_1 = 0, z_2 = 0$), fixed length $r_1 = L_1$ and $|\vec{r}_2 - \vec{r}_1| = L_2$. Total independent angles θ_1, θ_2 . Hence DOF = 2.
2. **Rigid Dumbbell in 3D:** $N = 2$ particles in 3D (6 coordinates) subject to $k = 1$ constraint ($|\vec{r}_1 - \vec{r}_2| = d$). Hence DOF = $3(2) - 1 = 5$ (3 translational, 2 rotational).

2. Vector Calculus and Potential Energy Function

2.1 Gradient, Divergence, and Curl

Definition: Gradient, Divergence, and Curl

- **Gradient (∇V):** Vector representing the rate and direction of maximum spatial change of a scalar field $V(x, y, z)$.
- **Divergence ($\nabla \cdot \vec{A}$):** Scalar representing the net outward flux of a vector field per unit volume.
- **Curl ($\nabla \times \vec{A}$):** Vector representing the circulation or rotational density of a vector field.

Mathematical Definitions in Cartesian Coordinates

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \quad (6)$$

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (7)$$

$$\text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (8)$$

2.2 Potential Energy Function $F = -\text{grad } V$

For a conservative force field $\vec{F}$, the work done along a closed path is zero ($\oint \vec{F} \cdot d\vec{r} = 0$). This implies $\vec{F}$ can be expressed as the negative gradient of a scalar potential energy function $V(\vec{r})$:

$$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}\right) \quad (9)$$

Physical Meaning of Negative Sign: The force acts in the direction of *decreasing* potential energy (towards stable equilibrium).

Equipotential Surfaces: A surface on which the potential energy $V(\vec{r}) = \text{constant}$.

- Work done in moving a mass along an equipotential surface is zero: $dW = \vec{F} \cdot d\vec{r} = -dV = 0$.
- The force vector $\vec{F}$ is everywhere perpendicular to the equipotential surface passing through that point.

Solved Problem 3: Deriving Force from Potential Energy

Problem: Given potential energy $V(x, y, z) = 3x^2y - y^3z^2$, calculate force $\vec{F}$ at point $(1, -1, 2)$.

Solution:

1. Calculate partial derivatives: $\frac{\partial V}{\partial x} = 6xy$, $\frac{\partial V}{\partial y} = 3x^2 - 3y^2z^2$, $\frac{\partial V}{\partial z} = -2y^3z$.
2. Evaluate at $(1, -1, 2)$: $\frac{\partial V}{\partial x} = 6(1)(-1) = -6$, $\frac{\partial V}{\partial y} = 3(1)^2 - 3(-1)^2(2)^2 = 3 - 12 = -9$, $\frac{\partial V}{\partial z} = -2(-1)^3(2) = 4$.
3. Substitute into $\vec{F} = -\nabla V$:

$$\vec{F} = -(-6\hat{i} - 9\hat{j} + 4\hat{k}) = 6\hat{i} + 9\hat{j} - 4\hat{k} \text{ N} \quad (10)$$

3. Conservative Forces and Conservation Laws

3.1 Conservative vs. Non-Conservative Forces

Comparative Breakdown:

Property	Conservative Force	Non-Conservative Force
Work Dependence	Work done depends only on initial and final positions.	Work done depends on the actual path taken.
Closed Path Work	$\oint \vec{F} \cdot d\vec{r} = 0$.	$\oint \vec{F} \cdot d\vec{r} \neq 0$.
Curl Condition	$\nabla \times \vec{F} = 0$ (Irrotational).	$\nabla \times \vec{F} \neq 0$.
Potential Energy	Expressible as $\vec{F} = -\nabla V$.	Potential energy function cannot be defined.
Mechanical Energy	Total mechanical energy ($E = K + V$) is conserved.	Mechanical energy is dissipated as heat/sound.
Examples	Gravitational force, Electrostatic force, Spring force.	Friction, Viscosity, Air resistance.

3.2 Fundamental Conservation Laws

Conservation Theorems

1. **Conservation of Linear Momentum:** If net external force $\vec{F}_{\text{ext}} = 0$, then:

$$\frac{d\vec{P}}{dt} = 0 \implies \vec{P} = \sum m_i \vec{v}_i = \text{constant} \quad (11)$$

2. **Conservation of Angular Momentum:** If net external torque $\vec{\tau}_{\text{ext}} = \vec{r} \times \vec{F}_{\text{ext}} = 0$, then:

$$\frac{d\vec{L}}{dt} = 0 \implies \vec{L} = \vec{r} \times \vec{P} = I\vec{\omega} = \text{constant} \quad (12)$$

3. **Conservation of Energy:** In an isolated system with conservative forces:

$$E_{\text{total}} = K + V = \frac{1}{2}mv^2 + V(\vec{r}) = \text{constant} \quad (13)$$

4. Non-Inertial Frames of Reference

4.1 Inertial vs. Non-Inertial Frames

Definition: Frames of Reference

- **Inertial Frame:** A frame of reference moving with zero acceleration (at rest or constant velocity) in which Newton's laws of motion hold strictly.
- **Non-Inertial Frame:** An accelerating or rotating frame of reference in which Newton's laws do not hold directly unless fictitious / pseudo forces are introduced.

4.2 Rotating Frames: Centrifugal and Coriolis Forces

Consider a frame S' rotating with constant angular velocity $\vec{\omega}$ relative to a stationary inertial frame S .

Transformation Equation for Acceleration

The transformation relation between acceleration in fixed frame $\vec{a}_S$ and rotating frame $\vec{a}_{S'}$ is:

$$\vec{a}_S = \vec{a}_{S'} + 2(\vec{\omega} \times \vec{v}') + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (14)$$

Multiplying by mass m , the effective force in the rotating frame is:

$$\vec{F}' = m\vec{a}_{S'} = \vec{F}_{\text{real}} - \underbrace{2m(\vec{\omega} \times \vec{v}')}_{\text{Coriolis Force } \vec{F}_{\text{cor}}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centrifugal Force } \vec{F}_{\text{cent}}} \quad (15)$$

Properties of Rotating Fictitious Forces:

Force	Vector Expression	Key Characteristics
Centrifugal Force	$\vec{F}_{\text{cent}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}) = m\omega^2 r_{\perp} \hat{r}$	Directed radially outward from axis of rotation. Present even if particle is stationary in rotating frame ($\vec{v}' = 0$).
Coriolis Force	$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$	Perpendicular to both velocity $\vec{v}'$ and angular velocity $\vec{\omega}$. Acts only when particle is in motion ($\vec{v}' \neq 0$). Does zero work ($W = 0$).

Physical Effects of Coriolis Force on Earth:

- **Deflection of Moving Objects:** Moving bodies in Northern Hemisphere are deflected to their *right*, while in Southern Hemisphere to their *left* (causes cyclone rotation direction).
- **Eastward Deflection of Freely Falling Body:** A body dropped from height h at latitude λ experiences an eastward deflection $\Delta x = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}}$.
- **Foucault Pendulum:** Demonstrates Earth's rotation directly via precession of its oscillation plane at angular rate $\Omega = \omega \sin \lambda$.

Solved Problem 4: Eastward Deflection Calculation

Problem: Calculate the eastward deflection due to Coriolis force for a body dropped from a height $h = 100$ m at Kolkata (latitude $\lambda = 22.5^\circ\text{N}$). (Take $\omega = 7.29 \times 10^{-5}$ rad/s, $g = 9.8$ m/s²).

Solution:

1. Formula for eastward deflection:

$$d = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}} \quad (16)$$

2. Calculate time of fall: $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{200}{9.8}} = 4.518$ s.

3. Substitute values: $d = \frac{1}{3}(7.29 \times 10^{-5})(9.8) \cos(22.5^\circ)(4.518)^3 \cos 22.5^\circ = 0.9239$, $(4.518)^3 = 92.23$. $d = 0.333 \times 7.29 \times 10^{-5} \times 9.8 \times 0.9239 \times 92.23 = \mathbf{0.0195 \text{ m} = 1.95 \text{ cm}}$.

5. Harmonic Oscillations

5.1 Damped Harmonic Motion

When an oscillator experiences a viscous damping force proportional to velocity ($F_{\text{damping}} = -b\dot{x}$), the equation of motion is:

$$m\ddot{x} + b\dot{x} + kx = 0 \implies \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0 \quad (17)$$

where $\gamma = \frac{b}{2m}$ is the damping coefficient and $\omega_0 = \sqrt{\frac{k}{m}}$ is the natural frequency.

Assuming solution $x(t) = Ce^{rt}$, the roots of characteristic equation are $r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$.

Three Cases of Damped Motion

1. **Overdamped** ($\gamma > \omega_0$): Non-oscillatory exponential return to equilibrium.

$$x(t) = e^{-\gamma t} \left(Ae^{\sqrt{\gamma^2 - \omega_0^2} t} + Be^{-\sqrt{\gamma^2 - \omega_0^2} t} \right) \quad (18)$$

2. **Critically Damped** ($\gamma = \omega_0$): Fastest non-oscillatory return to equilibrium without overshoot. Used in shock absorbers and deadbeat galvanometers.

$$x(t) = (A + Bt)e^{-\gamma t} \quad (19)$$

3. **Underdamped** ($\gamma < \omega_0$): Oscillatory motion with exponentially decaying amplitude.

$$x(t) = A_0 e^{-\gamma t} \cos(\omega_d t + \phi), \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \gamma^2} \quad (20)$$

Damping Parameters: Logarithmic Decrement and Quality Factor

- **Relaxation Time** (τ): Time in which energy drops to $1/e$ of initial value: $\tau = \frac{1}{2\gamma} = \frac{m}{b}$.
- **Logarithmic Decrement** (δ): Natural logarithm of ratio of successive amplitudes one period T_d apart:

$$\delta = \ln \left(\frac{A_n}{A_{n+1}} \right) = \gamma T_d = \frac{b T_d}{2m} \quad (21)$$

- **Quality Factor** (Q): Dimensionless measure of sharpness of oscillation energy retention:

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma} = \frac{\omega_0 \tau}{1} = \frac{\pi}{\delta} \quad (22)$$

5.2 Forced Oscillations and Resonance

Under a periodic external driving force $F(t) = F_0 \cos(\omega t)$, the differential equation is:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad \left(\text{where } f_0 = \frac{F_0}{m} \right) \quad (23)$$

In steady state, the response oscillates at driving frequency ω :

$$x(t) = x_0 \cos(\omega t - \phi) \quad (24)$$

where steady-state amplitude x_0 and phase lag ϕ are given by:

$$x_0(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \quad \tan \phi = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \quad (25)$$

Resonance Conditions

- **Amplitude Resonance Frequency (ω_r):** Frequency at which displacement amplitude x_0 is maximum:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}, \quad x_{0,\max} = \frac{f_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{f_0}{2\gamma\omega_0} \quad (\text{for small } \gamma) \quad (26)$$

- **Velocity Resonance Frequency (ω_v):** Frequency at which velocity amplitude $v_0 = \omega x_0$ is maximum:

$$\omega_v = \omega_0 \quad (\text{Natural undamped frequency}) \quad (27)$$

- **Phase at Resonance:** At velocity resonance ($\omega = \omega_0$), $\tan \phi = \infty \Rightarrow \phi = 90^\circ$ (force and velocity are in phase).

6. Rigid Body Dynamics and Moment of Inertia

6.1 Rigid Body Motion and Angular Velocity Vector

A rigid body is a system of particles in which inter-particle distances remain constant under external forces ($|\vec{r}_i - \vec{r}_j| = \text{constant}$).

General motion of a rigid body is a combination of translation of center of mass (CM) and rotation about CM. The angular velocity vector $\vec{\omega}$ is a vector directed along the instantaneous axis of rotation according to the right-hand rule.

Linear velocity of any particle i at position $\vec{r}_i$ relative to an axis is:

$$\vec{v}_i = \vec{\omega} \times \vec{r}_i \quad (28)$$

6.2 Moment of Inertia and Parallel/Perpendicular Axes Theorems

Definition: Moment of Inertia

Moment of Inertia (I) measures the rotational inertia of a body about a given axis. For a continuous mass distribution:

$$I = \sum m_i r_i^2 = \int r^2 dm = mk^2 \quad (29)$$

where k is the **radius of gyration**.

Fundamental Theorems of Moment of Inertia

1. **Parallel Axis Theorem:** MI about any axis is equal to MI about a parallel axis through CM plus Mh^2 :

$$I = I_{\text{cm}} + Mh^2 \quad (h = \text{perpendicular distance between parallel axes}) \quad (30)$$

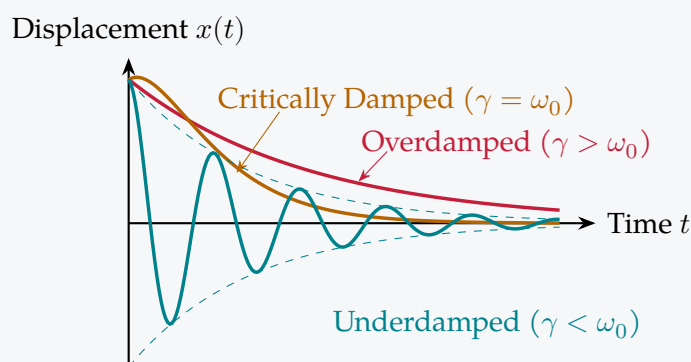
2. **Perpendicular Axis Theorem:** For a planar lamina in xy -plane, MI about z -axis perpendicular to lamina is:

$$I_z = I_x + I_y \quad (31)$$

Standard Moments of Inertia Summary:

Body	Axis of Rotation	Moment of Inertia (I)
Thin Uniform Rod (Length L)	Perpendicular through center.	$I = \frac{1}{12}ML^2$
Thin Uniform Rod (Length L)	Perpendicular through one end.	$I = \frac{1}{3}ML^2$
Circular Ring (Radius R)	Central axis perpendicular to plane.	$I = MR^2$
Circular Disc (Radius R)	Central axis perpendicular to plane.	$I = \frac{1}{2}MR^2$
Solid Sphere (Radius R)	Any diameter axis.	$I = \frac{2}{5}MR^2$
Spherical Shell (Radius R)	Any diameter axis.	$I = \frac{2}{3}MR^2$
Solid Cylinder (Radius R)	Central longitudinal axis.	$I = \frac{1}{2}MR^2$

Damped Harmonic Oscillator Response Regimes



Quick Revision — Module 1

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Key Formulas and Metrics at a Glance

Degrees of Freedom	$f = 3N - k$ (N particles, k constraints)
Limiting Static Friction	$f_{s,\max} = \mu_s N$ and $\tan \lambda = \mu_s$
Potential Energy Relation	$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$
Conservative Force Rule	$\nabla \times \vec{F} = 0 \iff \oint \vec{F} \cdot d\vec{r} = 0$
Coriolis Acceleration	$\vec{a}_{\text{cor}} = 2(\vec{\omega} \times \vec{v}')$ (Deflection to right in Northern Hemisphere)
Underdamped Frequency	$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ ($\gamma = b/2m$)
Quality Factor	$Q = \frac{\omega_0}{2\gamma} = \frac{\pi}{\delta} = \omega_0 \tau$
Resonance Amplitude Freq.	$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$
Parallel Axis Theorem	$I = I_{\text{cm}} + Mh^2$
Perpendicular Axis Theorem	$I_z = I_x + I_y$ (Planar lamina in xy -plane)

One-Line Definitions

- **Holonomic Constraint:** A constraint expressible as an integrable algebraic equation of position coordinates and time $f(\vec{r}, t) = 0$.
- **Conservative Force:** A force field in which work done along any closed path is zero ($\oint \vec{F} \cdot d\vec{r} = 0$).
- **Coriolis Force:** A velocity-dependent fictitious force acting in a rotating reference frame ($\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$).
- **Logarithmic Decrement:** The natural log of ratio of two consecutive amplitudes one period apart ($\delta = \gamma T_d$).
- **Radius of Gyration:** The perpendicular distance from axis of rotation where whole mass can be assumed concentrated ($I = Mk^2$).

Frequently Asked / Viva Questions

Q1. State the condition for a force field to be conservative.

A force field $\vec{F}$ is conservative if $\nabla \times \vec{F} = 0$ (irrotational) or equivalently work done around any closed loop $\oint \vec{F} \cdot d\vec{r} = 0$.

Q2. Why is Coriolis force called a fictitious force?

Because it arises due to the acceleration of the observer's rotating frame of

Frequently Asked / Viva Questions (contd.)

reference, not due to physical interactions between bodies.

Q3. What is the physical significance of negative gradient of potential energy?

$\vec{F} = -\nabla V$ indicates force always points in direction of steepest decrease of potential energy towards stable equilibrium.

Q4. Distinguish between holonomic and non-holonomic constraints with examples.

Holonomic constraints are integrable algebraic position relations (e.g. rigid body length); non-holonomic involve non-integrable velocity relations or inequalities (e.g. sphere rolling without slipping).

Q5. What is critical damping and state its practical application?

Critical damping occurs when $\gamma = \omega_0$. The system returns to equilibrium in shortest time without oscillating. Used in car shock absorbers and deadbeat galvanometers.

Q6. How does amplitude resonance frequency differ from velocity resonance frequency?

Amplitude resonance occurs at $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$ while velocity resonance occurs exactly at natural frequency $\omega_v = \omega_0$.

Q7. What is Quality Factor (Q) of an oscillator?

$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma}$. Higher Q implies lower damping and sharper resonance.

Q8. State Parallel and Perpendicular axes theorems of moment of inertia.

Parallel axis: $I = I_{\text{cm}} + Mh^2$. Perpendicular axis: $I_z = I_x + I_y$ for planar laminas.

Q9. Why are track rails laid higher on outer curves?

To provide necessary centripetal force via component of normal reaction and eliminate reliance on lateral friction.

Q10. Calculate the moment of inertia of a solid sphere of mass M and radius R about a tangent.

By parallel axis theorem, $I_{\text{tangent}} = I_{\text{cm}} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$.

Q11. What is the total energy of a simple harmonic oscillator of mass m and amplitude A ?

$E = \frac{1}{2}m\omega_0^2 A^2$. Total mechanical energy is constant and continuously converts between kinetic and potential forms.

Q12. Show that Coriolis force does no work on a moving particle.

$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$ is always perpendicular to velocity $\vec{v}'$. Power $P = \vec{F}_{\text{cor}} \cdot \vec{v}' = 0 \Rightarrow W = 0$.

Summary Comparison of Damped Motion Regimes

Parameter	Overdamped ($\gamma > \omega_0$)	Critically Damped ($\gamma = \omega_0$)	Underdamped ($\gamma < \omega_0$)
Roots of Char. Eq.	Real and distinct.	Real and equal ($r = -\gamma$).	Complex conjugate ($-\gamma \pm i\omega_d$).
Motion Nature	Non-oscillatory, slow return.	Non-oscillatory, fastest return.	Oscillatory with decaying amplitude.
Frequency	No frequency of oscillation.	No frequency of oscillation.	$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$.
Practical Use	Heavy door closers.	Galvanometers, vehicle suspension.	Tuning circuits, musical strings.